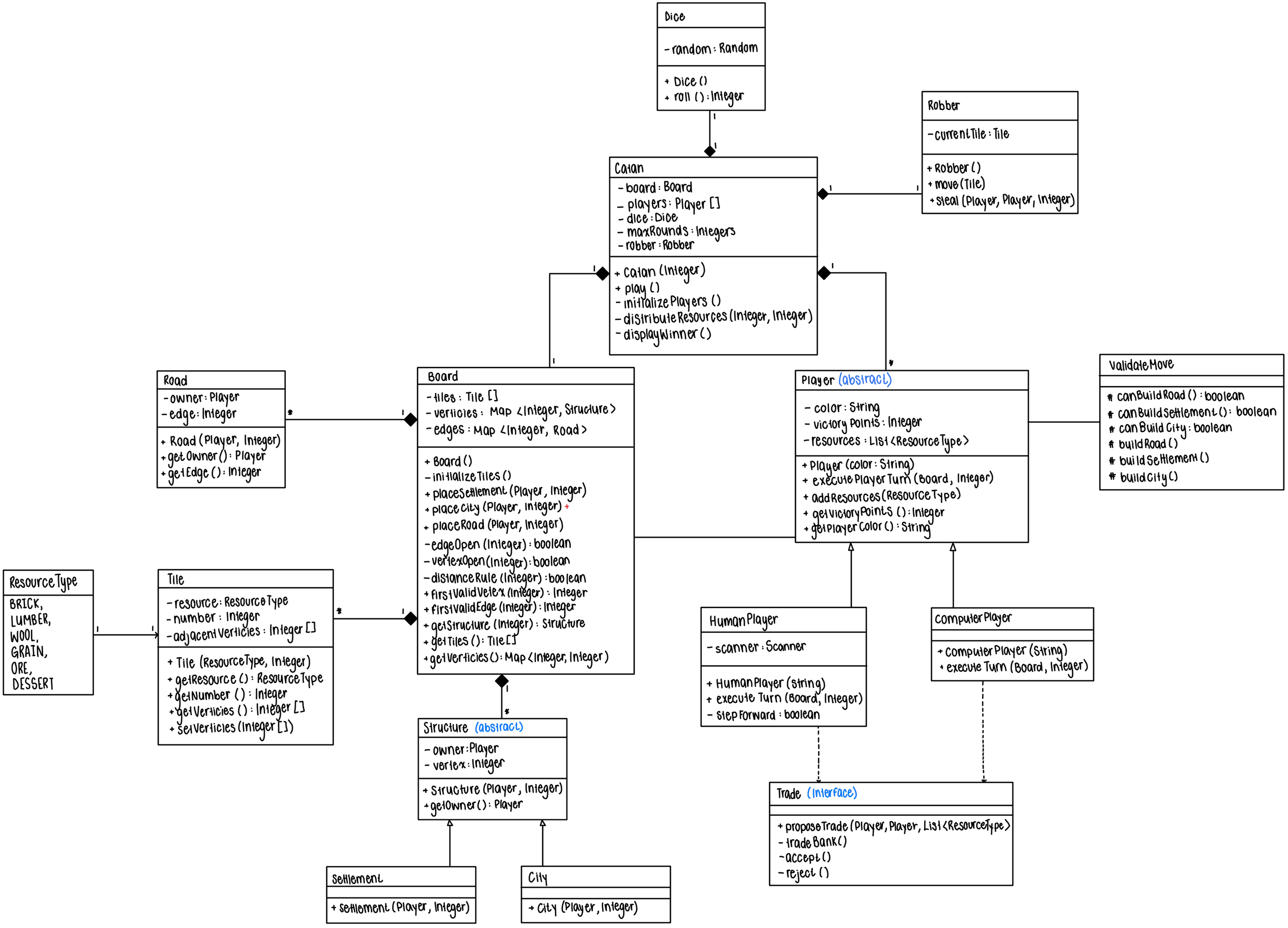
Dice
- random : Random
+ Dice ()
+ roll () : Integer
Robber
- currentTile : Tile
+ Robber ()
+ move (Tile)
+ steal (Player, Player, Integer)
Catan
- board : Board
- players : Player []
- dice : Dice
- maxRounds : Integers
- robber : Robber
+ Catan (Integer)
+ play ()
- initializePlayers ()
- distributeResources (Integer, Integer)
- displayWinner ()
Road
- owner : Player
- edge : Integer
+ Road (Player, Integer)
+ getOwner () : Player
+ getEdge () : Integer
Board
- tiles : Tile []
- verticies : Map <Integer, Structure>
- edges : Map <Integer, Road>
+ Board ()
- initializeTiles ()
+ placeSettlement (Player, Integer)
+ placeCity (Player, Integer)
+ placeRoad (Player, Integer)
- edgeOpen (Integer) : boolean
- vertexOpen (Integer) : boolean
- distanceRule (Integer) : boolean
+ firstValidVertex (Integer) : Integer
+ firstValidEdge (Integer) : Integer
+ getStructure (Integer) : Structure
+ getTiles () : Tile []
+ getVerticies () : Map <Integer, Integer>
Player (abstract)
- color : String
- victoryPoints : Integer
- resources : List <ResourceType>
+ Player (color : String)
+ executePlayerTurn (Board, Integer)
+ addResources (ResourceType)
+ getVictoryPoints () : Integer
+ getPlayerColor () : String
ValidateMove
# canBuildRoad () : boolean
# canBuildSettlement () : boolean
# canBuildCity : boolean
# buildRoad ()
# buildSettlement ()
# buildCity ()
ResourceType
BRICK,
LUMBER,
WOOL,
GRAIN,
ORE,
DESSERT
Tile
- resource : ResourceType
- number : Integer
- adjacentVerticies : Integer []
+ Tile (ResourceType, Integer)
+ getResource () : ResourceType
+ getNumber () : Integer
+ getVerticies () : Integer []
+ setVerticies (Integer [])
Structure (abstract)
- owner : Player
- vertex : Integer
+ Structure (Player, Integer)
+ getOwner () : Player
Settlement
+ settlement (Player, Integer)
City
+ City (Player, Integer)
HumanPlayer
- scanner : Scanner
+ HumanPlayer (String)
+ execute Turn (Board, Integer)
- stepForward : boolean
ComputerPlayer
+ ComputerPlayer (String)
+ execute Turn (Board, Integer)
Trade (Interface)
+ proposeTrade (Player, Player, List <ResourceType>)
- tradeBank()
- accept ()
- reject ()